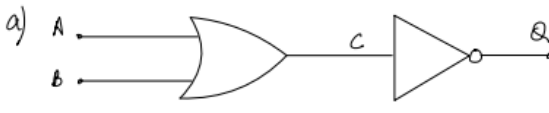
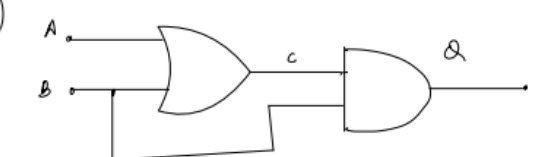


Tutorial 10 Questions (ELL101)

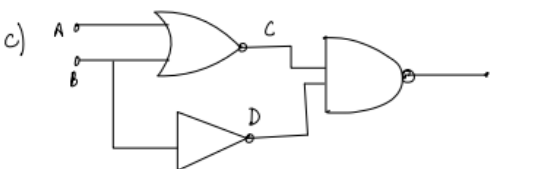
1. Convert the following binary number to its decimal equivalent ()₁₀
 - a) $(1011.101)_2$
 - b) $(1001.001)_2$
 - c) $(1101001011)_2$
2. Convert the following decimal numbers to its binary equivalent ()₂
 - a) 485
 - b) 451
3. What is the largest decimal number that can be represented using ten binary bits?
4. Add the following numbers in binary form. Verify by converting the numbers to decimal format and adding them.
 - (a) $101.11 + 11.01$
 - (b) $10.101 + 1.011$
 - (c) $1000101 + 1011001$
5. Represent -8 in 2's complement binary representation in 8-bit format.
6. The 2's complement form of a binary number is 10001. What is the number in decimal form?
7. Perform the following computations in binary form using 2's complement notations and check that they concur with the decimal equivalents:
 - a) $10 + 5$ b) $10 - 5$ c) $-10 + 5$ d) $-10 - 5$
8. Complete the truth tables for:

a) 

A	B	C	Q
0	0		
0	1		
1	0		
1	1		

b) 

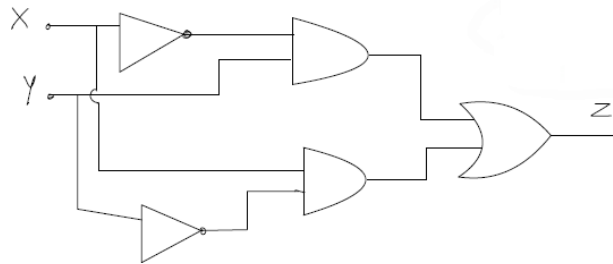
A	B	C	Q
0	0		
0	1		
1	0		
1	1		

c) 

A	B	C	Q
0	0		
0	1		
1	0		
1	1		

9. In the following circuit, X and Y are digital inputs and Z is digital output. The equivalent circuit is:

- a) NAND gate
- b) NOR gate
- c) XOR gate
- d) XNOR gate



10. The output of the following logic gate would be

- a) 0
- b) 1
- c) A
- d) $\bar{A}$

